

Building Risk Models to Predict Railroad Accident Severity

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Project Description

The purpose of this project is to investigate whether pre-incident features from the FRA Form 6180.54 dataset can help railroad industry professionals determine the severity of accidents/incidents involving their equipment. Feature engineering risk score and risk category as the two target variables for a regression and classification model.

Research Question: Is it possible to develop a risk model that helps professionals in the railroad industry determine the severity of an accident/ incident involving equipment?

Hypothesis: Pre-incident features from the FRA Form 6180.54 data source provide enough information to establish risk scores and classifications for railroad accidents more effectively than a dummy model.

Repository Structure

[Capstone Folder]

- | - Data # Raw Data, Data Dictionary, and Cleaned Data Sets

- | - cleaned_FRA_data_no_targets # *Cleaned Data Set*

- | - Raw Data and Dictionary

- | - FRA Raw Data # *Raw Dataset/ Used for FRA Reproduce Document*

- | - FRA (Form 54) Column Dictionary # *Feature Dictionary*

- | - Notebook Documents # *Jupyter Notebooks*

- | - 0) Capstone PostgreSQL Import Document # *Connecting Raw Data to Database*

- | - 1) Capstone Reproduce Document # *Document for Reproducibility*

- | - 2) Capstone Master Document # *Personal Document/ Containing SQL EDA*

- | - Presentation Documents #

- | - Capstone Final Presentation

- | - Capstone Final Report

Data Source

Data was pulled from the Federal Railroad Administration (FRA) Form 6180.54 via the publicly available U.S. Department of Transportation's site: <https://data.transportation.gov/Railroads/Rail-Equipment-Accident-Incident-Data-Form-54-/85tf-25kj>

The original dataset contained 224,700 records with 155 features, spanning from 1975 to 2026. Database used was PostgreSQL and pulled into a Jupyter Notebook. After data cleaning, the remaining records explored were 159,995 with 26 features.

Note: Data was having technical difficulties in reproducibility via using a CSV file versus a PostgreSQL pull. Corrections were made. Capstone Master Document was added to showcase EDA via SQL and to validate results if an error occurs.

How to Run

- 1) Download the Capstone Reproduce Document located in Notebook Document
- 2) Download data titled FRA Raw Data located in Data – Raw Data and Dictionary
- 3) Change pd.read_csv connection to data location on personal device
 - a. All packages needed are in the notebook
- 4) Run notebook – Takes roughly 10 minutes

Library Overview

- **Pandas:** [pandas - Python Data Analysis Library](#)
- **NumPy:** [NumPy](#)
- **Scikit-learn:** [scikit-learn: machine learning in Python — scikit-learn 1.9.0 documentation](#)
- **XGBoost:** https://xgboost.readthedocs.io/en/stable/python/python_api.html
- **SciPy:** [SciPy](#)
- **Matplotlib:** [Matplotlib — Visualization with Python](#)
- **Seaborn:** [seaborn: statistical data visualization — seaborn 0.13.2 documentation](#)

Methodology

- Data Split: Chronological - train (1975-1997)/ validation (1998-2011)/ test (2012-2026).
- Cross-validation: TimeSeriesSplit with 5 folds.
- Models: XGBoost Regressor and Classifier for final evaluation. DummyRegressor and DummyClassifier for baseline model. Linear Regression and KNN for exploration.
- Tuning: GridSearchSV – scoring via negative RMSE and F1-macro
- Statistical Tests: Paired t-test and bootstrap resample (5 Folds)

Results

- Regression Test: R-squared at 0.293 and RMSE at 0.934

- Regression Dummy: R-squared -0.002
- Classification Test: F1-macro at 0.61
- Classification Dummy: F1-macro at 0.503
- Models are statistically significant (p-value < 0.05 and 95% CI contain no 0 value)

Limitations

- Classification model misses a large number of High Severity Cases
- Linear Regression and KNN models were not evaluated for statistical testing against the XGBoost model
- Risk score is expressed in standard normal format, not optimal for non-technical users

Intended Use

This model is intended as a support tool for initial decision-making for mitigation strategy. Not a final decision-maker due to unexplained variance and missing High Severity cases. Additional investigation should be used.